

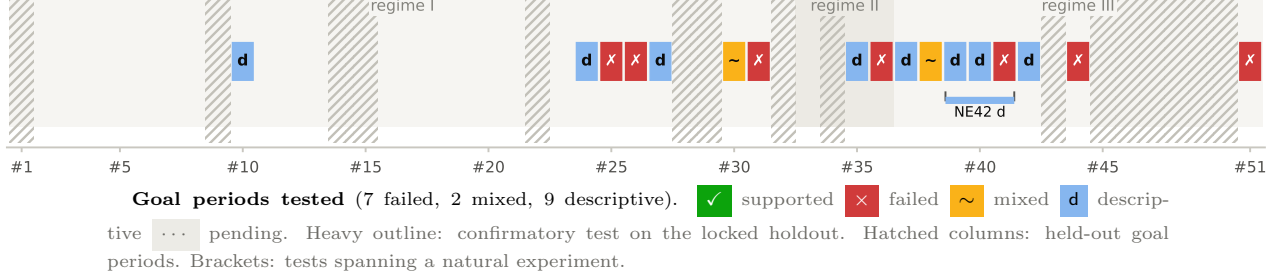
H68: Attention dilution is a mixture of two strategies

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1/5

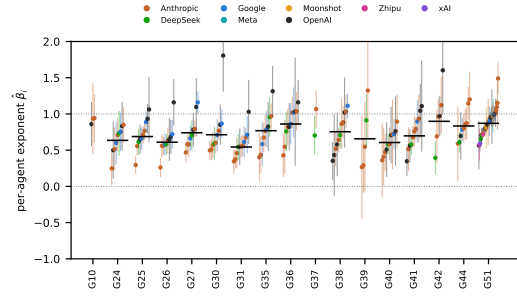
Bottom line. Every agent dilutes its attention with one narrow, unimodal exponent per period (between-agent SD 0.09–0.20), so $k^{-0.66}$ is a constant, not a mixture of two strategies.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH258 claims that the population dilution exponent ($k^{-0.66}$, H18) averages two kinds of agents: thin attenders ($\beta \approx 1$) and thread followers ($\beta \approx 0$), with lab-set weights. H68 fits one exponent per agent on H18's ledger pending sets in 17 periods (#10–#51, 170 agent-periods). A synthetic validation on the real skeletons came first. **The claim is refuted.** Agents scatter around the period mean with a between-agent SD of 0.09–0.20. The two-strategy world needs about 0.45. No period is bimodal (0/9 powered). In 7/9 powered periods the SD's upper 95% bound is below 0.35. No agent sits near $\beta = 0$ (lowest 0.25). Lab does not predict the exponent (p 0.37), and an agent's exponent barely persists across periods (ICC 0.23). Two natives: #51 fails, and NE42 has too few agents. The holdout script is written, not run.



Observables. Per-agent exponents $\hat{\beta}_i$ with 95% CIs, by period, colored by lab. The black bar is the random-effects mean. H68 predicts two clusters at 0 and 1 (dotted lines). Each period instead shows one cloud of width 0.1–0.2 around 0.55–0.87. No agent reaches 0, and the labs mix within every cloud.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 03 (contagion) with H18's uptake hazard. At talk turn τ of agent i , a pending sender j with n_j messages in a backlog of k is addressed with a cloglog hazard. $\theta_{i,d}$ is an agent-day propensity, γ a mention factor and β_i the agent's dilution exponent. The population layer is unimodal (U), the H68 two-delta mixture, or a free two-component mixture. Each is fitted on $\hat{\beta}_i$ with its own sampling variance s_i^2 .

$$P(r_{\tau j} = 1) = 1 - \exp[-\theta_{i,d} n_j k_{\tau}^{-\beta_i} e^{\gamma m_j}]$$

$$U: \hat{\beta}_i \sim \mathcal{N}(\mu, \tau^2 + s_i^2)$$

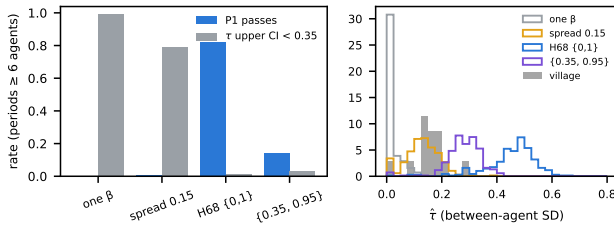
$$\text{H68: } \hat{\beta}_i \sim \pi \mathcal{N}(1, v_i) + (1 - \pi) \mathcal{N}(0, v_i), \quad v_i = \tau_w^2 + s_i^2$$

$$\text{LR} = 2[\ell_{M2} - \ell_U], \quad p \text{ by parametric bootstrap under } U$$

Dilution is an agent-level constant per period, not a property of the roster's family mix. A swarm designer can use one exponent per period ($\beta \approx 0.6$ –0.9) as the per-pair coupling normalization $J \propto N^{-\beta}$. A change of model families will not change it. There is no “thread-follower” species to protect from crowding: every agent ignores more of the backlog as the backlog grows. Concentrated agents dilute *more* ($\rho + 0.36$). To keep a message from being lost in a large room, address it or shrink the room. Changing which models sit in the room does not help. In practice this result is a negative control, and it adds no new lever.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 bimodal in $\geq 50\%$ powered	0/9; SD upper < 0.35 in 7/9; H68 fit 1–29 log-lik below U	failed
P2 lab $\eta^2 \geq 0.4$, $p < 0.05$; trait ≥ 0.5	$\eta^2 0.16$, $p 0.37$; ICC 0.23 $[-0.10, 0.50]$	failed
P3 mean of $\beta_i = \text{pooled } \beta$	15/15 within 0.035	passed
P4 $\rho(\beta, C) < -0.3$	+0.36 ($n 170$)	reversed
P5 mention vs reply rank ≥ 0.3	0.52 ($n 156$)	passed
#51: bimodal; timer wakes	lab; SD 0.15 $[0.11, 0.21]$; lab $p 0.12$; $\rho 0.15$	failed
NE42 trait merge	across the 5 agents; $\rho 0.40$ ($p 0.50$)	no power
Post hoc: H18's per-message mention factor	#51 mean 0.87 $\rightarrow 0.63$; all verdicts same	robust



Second figure. Synthetic validation on the real skeletons (17 periods, 4 worlds, 20 runs each). Left: in periods with at least 6 agents, the bimodality rule fires for the H68 world and never for one exponent. Right: the deconvolved spread $\hat{\tau}$. The village (gray) falls on the one-exponent and spread-0.15 worlds, far from H68's 0.45.

GOAL PERIODS IN DETAIL

G10 d descriptive
 G24 d descriptive
 G25 $\times$ failed
 G26 $\times$ failed
 G27 d descriptive
 G30 $\sim$ mixed
 G31 $\times$ failed
 G35 d descriptive
 G36 $\times$ failed
 G37 d descriptive
 +8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The response is H18's mention proxy. Reply labels keep one parent per message, so only their agent ranking is used. Eligibility drops 53 of 223 agent-periods with few talks, and a quiet thread follower could hide there. The per-agent timer-wake check (#51, 24 agents) does not exclude reactive timing per agent. The trait estimator is noisy. The level of β depends on the mention-factor convention, but the spread does not. The holdout run needs a pending-set builder that keeps held-out days.

ROUND 2

- **H68-R1** The residual spread ($\tau \approx 0.15$) is agent $\times$ period.
- **H68-R2** Fix one mention-factor convention across H18 and H68 (per message) in the shared pending-set builder.
- **H68-R3** Use one β per period as the mean-field normalization in H03/H05; no composition correction is needed.

H68-dilution-mixture · generated 2026-10-04 12:53 by infra/summaries/build-summaries.py · card hypotheses/H68-dilution-mixture/README.md · ratings: H68 agent (suggested) (page 2)